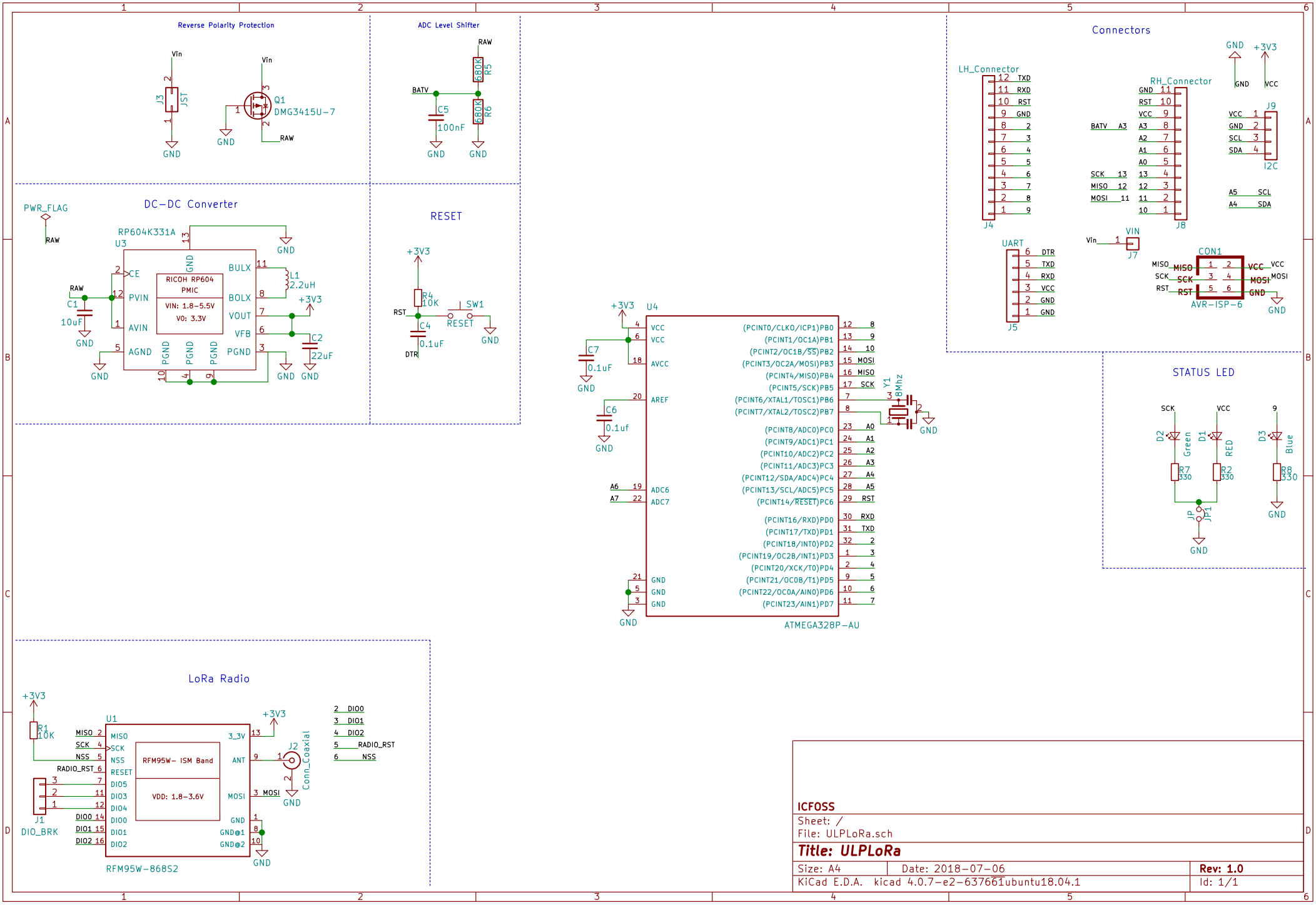
Reverse Polarity Protection
Vin
J3
JST
GND
Vin
Q1
DMG3415U-7
GND
RAW
ADC Level Shifter
RAW
680K
R5
BATV
C5
100nF
680K
R6
GND
GND
DC-DC Converter
PWR_FLAG
RAW
RP604K331A
U3
GND
RAW
C1
10uF
GND
CE
PVIN
AVIN
AGND
RICOH RP604
PMIC
VIN: 1.8-5.5V
VO: 3.3V
BULX
BOLX
VOUT
VFB
PGND
PGND
PGND
PGND
L1
2.2uH
+3V3
C2
22uF
GND
GND
GND
RESET
+3V3
R4
10K
RST
SW1
RESET
C4
0.1uF
DTR
GND
Connectors
LH_Connector
J4
TXD
RXD
RST
GND
RH_Connector
J8
BATV A3
SCK 13
MISO 12
MOSI 11
GND
RST
VCC
A3
A2
A1
A0
GND +3V3
GND VCC
J9
I2C
VCC
GND
SCL
SDA
A5 SCL
A4 SDA
UART
J5
DTR
TXD
RXD
VCC
GND
GND
VIN
Vin
J7
CON1
AVR-ISP-6
MISO
SCK
RST
VCC
MOSI
GND
GND
STATUS LED
SCK
VCC
9
D2
Green
D1
RED
D3
Blue
R7
330
R2
330
R8
330
JP
JP1
GND
GND
+3V3
U4
VCC
VCC
AVCC
AREF
C7
0.1uF
C6
0.1uf
GND
GND
Y1
8Mhz
GND
(PCINT0/CLKO/ICP1)PB0
(PCINT1/OC1A)PB1
(PCINT2/OC1B/SS)PB2
(PCINT3/OC2A/MOSI)PB3
(PCINT4/MISO)PB4
(PCINT5/SCK)PB5
(PCINT6/XTAL1/TOSC1)PB6
(PCINT7/XTAL2/TOSC2)PB7
(PCINT8/ADC0)PC0
(PCINT9/ADC1)PC1
(PCINT10/ADC2)PC2
(PCINT11/ADC3)PC3
(PCINT12/SDA/ADC4)PC4
(PCINT13/SCL/ADC5)PC5
(PCINT14/RESET)PC6
(PCINT16/RXD)PD0
(PCINT17/TXD)PD1
(PCINT18/INT0)PD2
(PCINT19/OC2B/INT1)PD3
(PCINT20/XCK/T0)PD4
(PCINT21/OC0B/T1)PD5
(PCINT22/OC0A/AIN0)PD6
(PCINT23/AIN1)PD7
MOSI
MISO
SCK
A0
A1
A2
A3
A4
A5
RST
RXD
TXD
ADC6
ADC7
GND
GND
GND
ATMEGA328P-AU
LoRa Radio
+3V3
R1
10K
U1
MISO
SCK
NSS
RESET
DIO5
DIO3
DIO4
DIO0
DIO1
DIO2
RFM95W- ISM Band
VDD: 1.8-3.6V
3_3V
ANT
MOSI
GND
GND@1
GND@2
+3V3
J2
Conn_Coaxial
GND
GND
J1
DIO_BRK
RADIO_RST
DIO0
DIO1
DIO2
RFM95W-868S2
2 DIO0
3 DIO1
4 DIO2
5 RADIO_RST
6 NSS
ICFOSS
Sheet: /
File: ULPLoRa.sch
Title: ULPLoRa
Size: A4
Date: 2018-07-06
Rev: 1.0
KiCad E.D.A. kicad 4.0.7-e2-637661ubuntu18.04.1
Id: 1/1